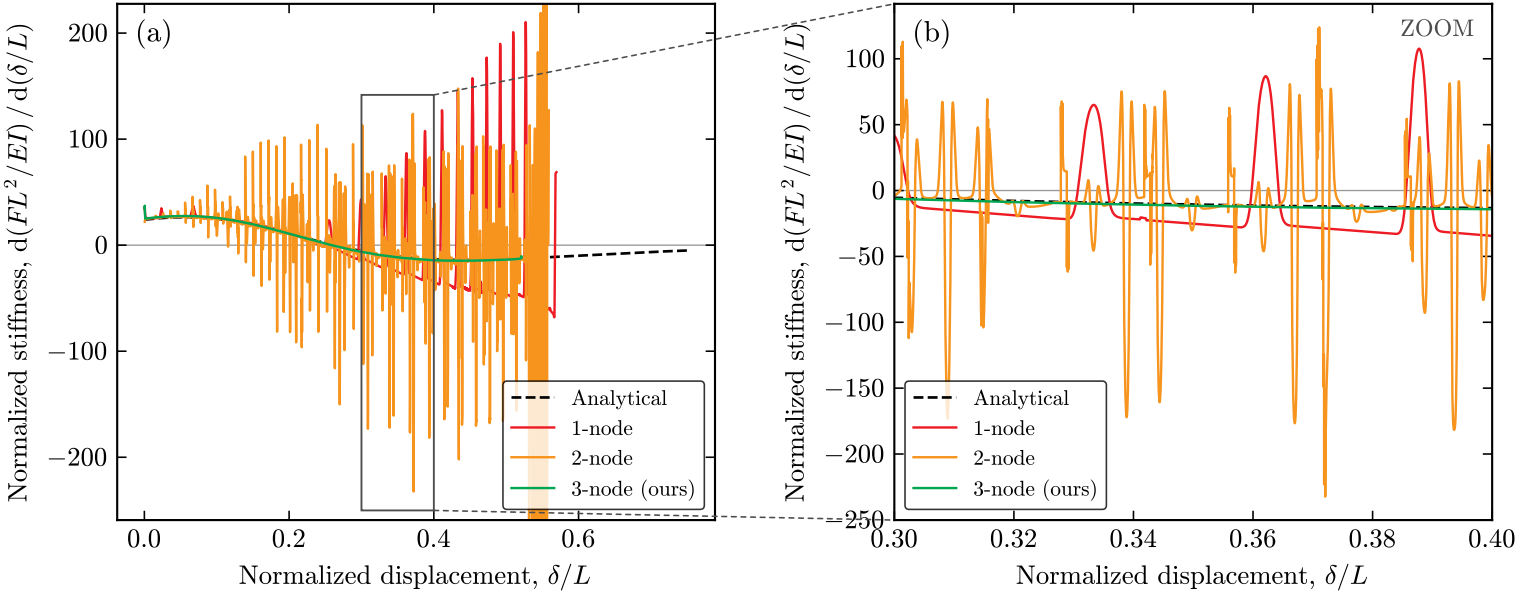
(a)
(b)
ZOOM
Analytical
1-node
2-node
3-node (ours)
Normalized stiffness, d(FL²/EI) / d(δ/L)
Normalized displacement, δ/L
200
100
0
−100
−200
0.0
0.2
0.4
0.6
50
−50
−150
−250
0.30
0.32
0.34
0.36
0.38
0.40